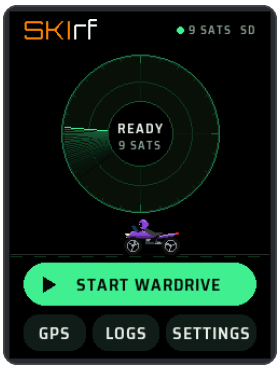
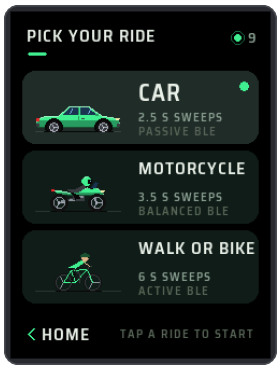


C5 CYD Edition
User Guide and Start-up Guide

NM-CYD-C5 (ESP32-C5)
Beta 1



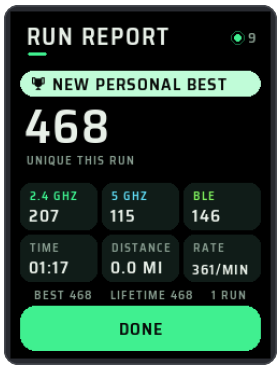
Home



Pick your ride



Wardrive



Run report

Printable guide. Screen pictures are example renderings of the interface; your screen will match the layout.

Contents

1. About this device 3

2. What you need 4

3. Quick start 5

4. Getting around 7

5. Going wardriving 9

6. Logs and uploads 11

7. The web portal 13

8. Settings reference 14

9. Pro mode 16

10. Care, safety and responsible use 17

11. Troubleshooting 18

Quick-reference card 19

Glossary 20

Appendix A. Technical summary 21

1. About this device

SKIrf Wardriver turns the NM-CYD-C5 (cheap yellow display) board into a pocket wardriver. With a GPS module fitted it listens for the Wi-Fi networks around you on **both 2.4 GHz and 5 GHz** and for **Bluetooth LE** devices, notes where you were when you heard each one, and saves the lot in the **WiGLE CSV** format that wardriving maps and leaderboards read. You can upload your runs to **WiGLE** and **WDGWars** straight from the board.

What it records

- **Wi-Fi networks** on 2.4 GHz and 5 GHz: network name (SSID), hardware address (BSSID), security, channel, frequency and signal strength.
- **Bluetooth LE devices**: address, name when it has one, manufacturer ID and signal strength.
- **Where and when**: latitude, longitude, altitude, accuracy and the GPS time for every row.

How it works in one minute

Every Wi-Fi access point sends a short **beacon** several times a second that says who it is, and many Bluetooth devices **advertise** themselves the same way. The board sweeps through every Wi-Fi channel on both bands and then listens for Bluetooth, over and over. Each new network or device is counted once per run and written to the log with your GPS position. If you later pass much closer to one (10 dB stronger), it is logged again so the map can place it better, but it is not counted twice.

How often it sweeps depends on the **ride** you pick: a car moves past networks quickly, so the Car profile sweeps every 2.5 seconds with short listening times; on foot it sweeps every 6 seconds and listens longer on every channel.

> **IMPORTANT** Read this once
>
> Wardriving records networks and devices that broadcast openly. It does not connect to them. Listening is lawful in many places, but laws differ and you are responsible for following the laws where you are. Never join networks you are not allowed to use, and do not use the logs to identify, follow or harass people. **Do not look at or touch the screen while driving.**

> **NOTE** About Beta 1
>
> This is a **beta**: the first release for general use. It does what this guide describes, but expect rough edges and some behaviour to change in later releases. Section 11 lists what to check if something does not behave as described, and the project page explains how to report a problem.

2. What you need

Item	Why	Notes
NM-CYD-C5 board	Runs the software	ESP32-C5, 2.8 inch touch screen, microSD slot, USB-C. 8 MB or 16 MB flash both work.
USB-C data cable	Flashing and power	Some cables only charge. If the computer does not see the board, try another cable.
GPS module	**Needed.** A run only scans with a fix	Any NMEA module at 9600 baud, for example an ATGM336H. Its TX goes to the board's IO4 by default (changeable in Settings > GPS pins).
microSD card	Recommended. Holds your logs and saved keys	FAT32, 32 GB or smaller. Fit it before switching on. Without a card, logs go to the board's flash (about 1.5 MB).
Power in the car	Long runs	A car USB port or a power bank. The board has no battery.
Passive buzzer	Optional: beeps	+ to IO26, - to GND. Without one the board is silent.
Phone or laptop	Optional: web portal	To download logs and save your Wi-Fi and API keys.
WiGLE / WDGWars accounts	Optional: uploads	Free accounts at wigle.net and wdgwars.pl.

Fitting the GPS

Connect the module's power and ground to the board, its **TX** to the board's **RX pin** (IO4) and, if it has one, its **RX** to the board's **TX pin** (IO5). The board uses the TX line to ask the module for position updates five times a second; a module that only sends NMEA works with the TX pin set to -1. Place the antenna where it can see the sky: on the dashboard, under the windscreen, or on top of a backpack.

> **TIP** Different wiring?
>
> If your module is on other pins, open Settings > GPS pins, pick a preset (4 / 5, 4 / -1, 3 / -1, 8 / 9) or set RX and TX with - and +, then tap SAVE. The board restarts with the new pins.

3. Quick start

3.1 Get the firmware file

Download **SKIrf_WARDRIVER_C5_CYD_Beta_1.bin** and **SHA256SUMS.txt** from the Releases page of the project (github.com/SKIrf26/WARDRIVER). To check the download, compare its checksum with the line in SHA256SUMS.txt (optional but recommended):

Computer	Command
Windows (PowerShell)	`Get-FileHash .\SKIrf_WARDRIVER_C5_CYD_Beta_1.bin -Algorithm SHA256`
macOS or Linux	`shasum -a 256 -c SHA256SUMS.txt`

3.2 Flash it to the board

1. Install **Python 3** if you do not have it, then the flashing tool: **pip install esptool** (on Windows, **py -m pip install esptool**).
2. Plug the board into the computer with the USB-C cable, using the **left** USB-C port.
3. Find the port: on **Windows** it is a COM port in Device Manager (for example COM5); on **macOS** /dev/cu.usbserial... (list with ls /dev/cu.*); on **Linux** /dev/ttyUSB0 or /dev/ttyACM0.
4. In a terminal, in the folder with the file, run (replace YOUR_PORT):

```
esptool --chip esp32c5 --port YOUR_PORT --baud 460800 write-flash 0x0
SKIrf_WARDRIVER_C5_CYD_Beta_1.bin
```

1. Wait for **Hash of data verified** and **Hard resetting**. The board restarts by itself.

> **TIP** If the computer cannot connect to the board
>
> Use a USB-C **data** cable and try another USB port. Then unplug the cable, **hold the board's BOOT button**, plug the cable back in, release BOOT and run the command again. If the connection is unreliable, add **--baud 115200**. The middle USB-C port also works, but you must hold BOOT, tap RESET and release BOOT yourself.

> **NOTE** What flashing changes
>
> Writing the file replaces the software and **resets the settings saved in the board**: touch calibration, theme, brightness, orientation, GPS pins, Pro mode and your records. Files on the microSD card, including your logs and saved Wi-Fi and API keys, are not touched.

3.3 First start

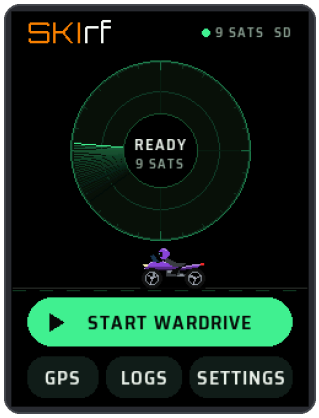
1. **Touch calibration.** The screen says TOUCH CAL, "Tap the four corners". A marker appears in one corner at a time: tap each exactly (a stylus or fingernail tip works best). The board says "Cal saved" and restarts.
2. **Boot screen.** Two vehicles drive past the SKIrf logo; **BETA 1** is shown under it.
3. **Web portal.** For 60 seconds the board offers its own Wi-Fi hotspot for a phone or laptop (section 7). Tap **SKIP** if you do not need it now.
4. **Home.** You land on the radar with **START WARDRIVE** and **GPS**, **LOGS** and **SETTINGS** below.
5. **Check the GPS.** Tap GPS. Outdoors, SATELLITES climbs and FIX changes from SEARCH to LOCKED (or 3D LOCK with a precise fix). The first fix after power-up can take a few minutes.



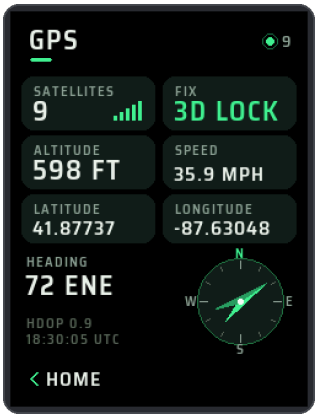
Boot screen



Web portal (60 s)



Home



GPS

4. Getting around

Everything starts at **Home**. Each screen has a title at the top left and a **BACK** or **HOME** button at the bottom left.

Screen	How to get there	What it is for
Home	Start-up, or HOME on any screen	Radar, the START WARDRIVE button, GPS, LOGS, SETTINGS
Pick your ride	Home > START WARDRIVE	Choose Car, Moto or Walk/Bike
Wardrive	Tap a ride	START and STOP a run, live counters
Run report	After STOP	Totals for the run, personal best, lifetime totals
GPS	Home > GPS	Satellites, fix, position, speed, heading, compass
Logs	Home > LOGS, or Settings > Logs	Select runs, upload to WiGLE or WDGWars, delete
Settings	Home > SETTINGS	Display, connections, your data, advanced (section 8)

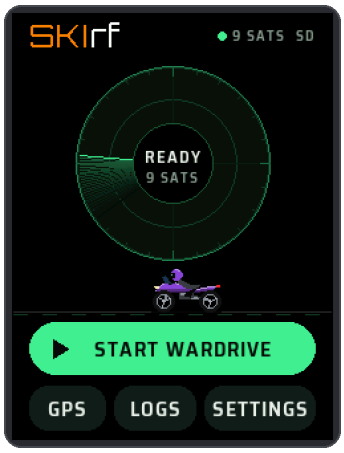
Touch gestures

Do this	To get this
Tap	Press a button, pick a row. The screen is resistive: a firm tap with a fingertip, fingernail or stylus works best.
Drag up or down on a list	Scroll the list (Settings, Theme, Logs, Choose network, About, WDGWars stats, Pro). A slim bar on the right shows where you are. A row is only picked when you lift your finger without dragging.
Tap STOP, then TAP TO STOP	End a run. The second tap must come within 3 seconds, so a bump will not end your run.
Press and hold an empty spot on Home for 3 s	Start touch calibration again
Hold BOOT while plugging in	Start touch calibration again when touch is too far off to reach Settings

The status at the top right

Most screens show the GPS status at the top right: a dot with a ring and the number of satellites in use. The dot is **green** when the GPS has a fix, **pale** when it sees satellites but has no fix yet, and **dim** when it hears nothing. While a run is recording, a **red dot blinks** beside it. Home shows the satellites and where logs go (**SD** for the card).

Home



Home

The **radar** sweeps all the time. While you record, every new network or device lands as a blip: the closer to the middle, the stronger the signal, and the colour shows the band (2.4 GHz, 5 GHz or Bluetooth). Blips fade after a while. The middle shows READY and the satellites, or the live total while recording.

Your **ride** idles on the road below the radar. **START WARDRIVE** opens the ride picker; while a run is recording it takes you back to the dashboard.

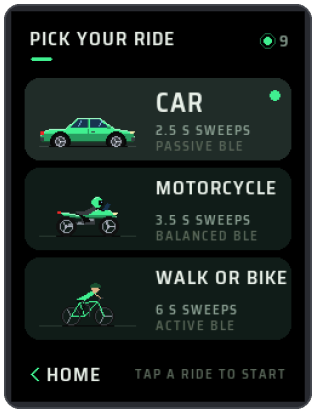
GPS, **LOGS** and **SETTINGS** open those screens.

5. Going wardriving

5.1 Before you set off

- Fit the microSD card before switching on (Settings shows **SD card** next to Logs when it was found).
- Put the GPS antenna where it can see the sky and wait for a fix (the status dot turns green).
- Mount the board where a passenger can see it, or where you can glance at it when parked. Tap START before you drive off.

5.2 Pick your ride



Pick your ride

Tap **START WARDRIVE** on Home, then tap a ride. The ride sets how the board scans:

Ride	Sweep	Scanning
CAR	every 2.5 s	Short listening time per channel, passive Bluetooth. The slow 5 GHz radar (DFS) channels are skipped on two sweeps in three.
MOTO	every 3.5 s	A little longer per channel, balanced Bluetooth, same DFS rule as Car.
WALK / BIKE	every 6 s	Longest listening time, active Bluetooth, every channel on every sweep.

Each ride needs a GPS fix with at least **4 satellites** (5 for Walk/Bike) before a sweep starts. The ride also picks the vehicle on screen and the letter in the log file name (C, M or W).

5.3 The wardrive screen



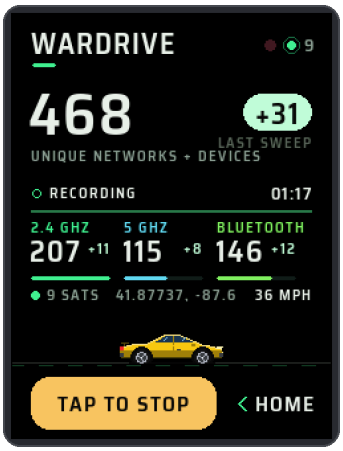
Recording

Tap **START**. The board beeps twice, opens a new log file and starts sweeping. Your ride drives along while you record and parks when you stop.

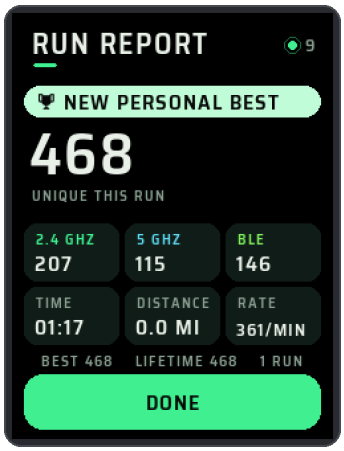
Part	What it shows
Big number	Unique networks and devices caught this run
+N badge	New catches in the last sweep
Status line	RECORDING and the run time, or why it is waiting (Waiting for a GPS fix, Waiting for more satellites)
2.4 GHZ / 5 GHZ / BLUETOOTH	Unique count per band, new in the last sweep (+N), and a bar for each band's share
GPS strip	Satellites, position and speed

> **NOTE** Waiting for the GPS
>
> While the GPS has no fix, or too few satellites, the radios stay idle and the status line says so. This keeps every row in your log on the map. Scanning starts by itself as soon as the fix is good enough.

5.4 Stopping and the run report



First tap: TAP TO STOP



Run report

Tap **STOP**: it turns amber and says **TAP TO STOP**. Tap again within 3 seconds to end the run; otherwise it goes back to STOP and the run carries on. The board writes the last rows, beeps, and shows the **Run report**:

- the unique total for the run, split into 2.4 GHz, 5 GHz and BLE,
- run time, distance travelled (from the GPS) and catch rate per minute,
- **NEW PERSONAL BEST** when you beat your best run, plus your best, lifetime total and number of runs.

Tap **DONE** to return. Your records are kept until you flash a new image.

5.5 Tips for big numbers

- Pick the ride that matches your speed. Car on foot wastes time between sweeps; Walk/Bike in a car misses networks while it listens.
- Busy streets, apartment blocks and shopping areas give the most networks per minute. Slow traffic helps.
- Keep the board and GPS antenna high and near a window. Metal and tinted glass cut the signal.
- Use a microSD card: the board's own flash fills up after a few long runs.
- Upload after each session so your WiGLE and WDGWars totals stay current.

6. Logs and uploads

6.1 The log files

Every run makes a new file in the top folder of the microSD card (or the board's flash when there is no card). The name tells you the ride and the GPS date and time the run started:

```
SKIrf_WD_C_20261017_183005.csv
```

C is Car, M is Moto and W is Walk/Bike. A file is never added to: each run gets its own. The format is **WiGLE CSV 1.6** (Appendix B), which WiGLE and WDGWars accept as it is and any spreadsheet can open.

6.2 The Logs screen



Logs

Home > **LOGS** lists your runs with an icon for the ride. The line at the top says how many there are and how many are selected. Drag to scroll.

Button	What it does
Tap a run	Select or unselect it (a tick appears)
ALL / NONE	Select every run, or clear the selection
WIGLE	Upload the selected runs to WiGLE (6.3)
WDGW	Upload the selected runs to WDGWars (6.4)
DELETE	Delete the selected runs: tap DELETE, then CONFIRM
BACK	Return

> **NOTE** Uploads need your home Wi-Fi
>
> To upload, the board joins your home Wi-Fi (6.5) and needs your account keys, saved once in the web portal (section 7). It cannot upload while a run is recording. A screen shows each file as it goes and the result at the end.

6.3 WiGLE

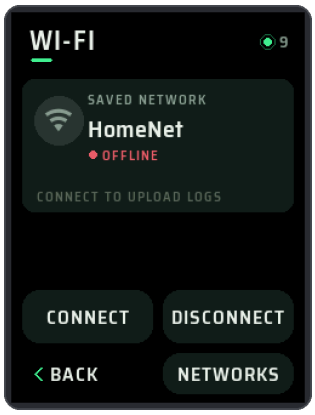
1. Make a free account at **wigle.net**. Open your **Account** page and find the **API Name** and **API Token** (these are not your login name and password).
2. Open the web portal (section 7), tap **WIGLE API KEY**, paste both, tap **SAVE CREDENTIALS**.
3. On the board: LOGS, select runs, **WIGLE**. Or in the portal: tick runs and **UPLOAD SELECTED** → **WIGLE**.

6.4 WDGWars

WDGWars (wdgwars.pl) is a wardriving game with gangs and leaderboards.

1. Copy the **API key** from your WDGWars profile.
2. In the web portal tap **WDGWARS API KEY**, paste it, tap **SAVE KEY**.
3. On the board: LOGS, select runs, **WDGW**.
4. Settings > **WDGWars stats** loads your totals: player, gang, Wi-Fi, Bluetooth, aircraft and mesh counts, total, today and the last 7 days. **REFRESH** loads them again.

6.5 Home Wi-Fi



Settings > Wi-Fi

Settings > **Wi-Fi** shows the saved network and whether the board is connected.

Button	What it does
NETWORKS	Scan and list nearby networks with their signal. Tap one, type the password on the big-key keyboard, tap JOIN.
CONNECT	Join the saved network now
DISCONNECT	Leave it and stop reconnecting

You can also save the network in the web portal (HOME WIFI SETUP). While the board is connected to your home Wi-Fi, the web portal page is also on your home network at the address shown here.

7. The web portal



Web portal

For **60 seconds** after each start, and whenever you choose Settings > **Web portal**, the board makes its own Wi-Fi hotspot. The ring counts down; **SKIP** closes it early.

1. On a phone or laptop, join the Wi-Fi called **SKIrf_WARDRIVER** (no password).

2. Open a browser at **http://192.168.4.1** (not https).

The page lists the logs on the card or flash with their sizes.

On the page	What it does
Tick boxes, select all	Choose runs for the buttons below
UPLOAD SELECTED → WIGLE / WDGWARS / ALL LOGGED-IN	Upload the ticked runs (the board joins your home Wi-Fi to do it)
DELETE SELECTED CSV	Delete the ticked runs (asks first)
A file name	Download that run
DOWNLOAD ALL .CSV (.ZIP)	Download every run in one .zip file
DELETE ALL CSV	Delete every run (asks first; cannot be undone)
HOME WIFI SETUP	Save your home network name and password
WIGLE API KEY / WDGWARS API KEY	Save your upload keys (6.3, 6.4)
Return to device	Close the portal and go back to Home

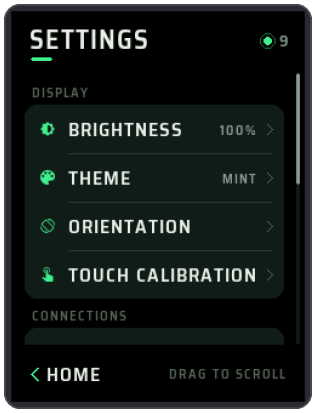
IMPORTANT The hotspot has no password

While the portal is up, anyone in range can join it and see, download or delete your logs. Tap SKIP when you do not need it and do not open it in a crowded place. Saved passwords and keys are never shown on the page: their fields say "saved - leave blank to keep".

TIP Phone keeps dropping the hotspot

Some phones leave a Wi-Fi network that has no internet. Turn off "Wi-Fi assist" or "switch to mobile data" while you use the portal, or use a laptop.

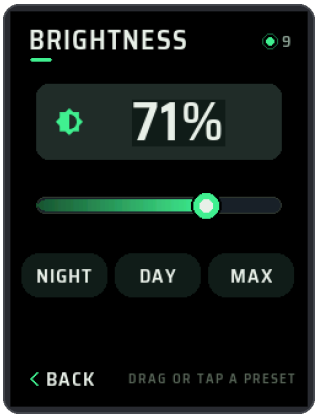
8. Settings reference



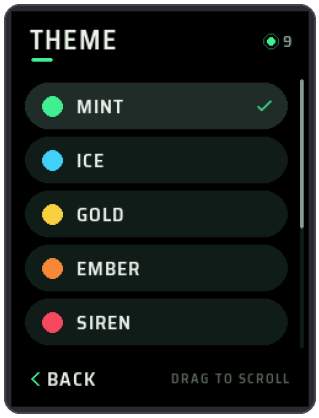
Settings

Home > **SETTINGS** is one list in four groups. Each row shows its current value on the right. Drag to scroll; tap a row to open it.

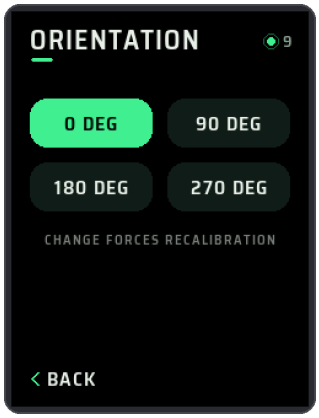
Setting	What it does
DISPLAY	
Brightness	Drag the slider, or tap NIGHT, DAY or MAX
Theme	Accent colour: Mint, Ice, Gold, Ember, Siren, Neon, Cobalt, Chrome, or WDGW (pink and cyan). All colours on screen are matched to it.
Orientation	0, 90, 180 or 270 degrees. A change asks for a new touch calibration.
Touch calibration	Tap the four corner markers again; the board restarts
CONNECTIONS	
Wi-Fi	Your home network for uploads (6.5)
Web portal	Open the 60-second portal (section 7)
GPS pins	RX and TX pins for the GPS module, with presets. SAVE restarts the board.
YOUR DATA	
Logs	The Logs screen (6.2). The value says SD card or Flash.
WDGWars stats	Your WDGWars totals (6.4)
ADVANCED	
Pro mode	The operator console (section 9)
About & safety	Photosensitivity notice and terms of use



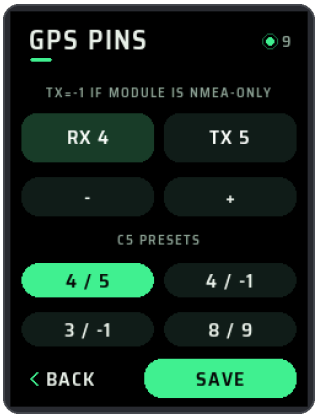
Brightness



Theme



Orientation



GPS pins

9. Pro mode



Pro mode

Pro mode is a console for people who want to tune the scanner by hand. It switches the screens to a plain green-on-black look and adds the settings below. Turn it on from Settings > **Pro mode**; tap a row to step through its values.

> **NOTE** Pro mode is slower by default
>
> In Pro mode the **Sweep interval** replaces the ride's own (3, 5, 8 or 12 s; 8 s at first), so it sweeps less often than the Car profile. Set it to 3 s for driving.

Row	Values	What it does
Wi-Fi scan	On / Off	Scan Wi-Fi
BLE scan	On / Off	Scan Bluetooth LE
BLE duty	Low / Bal / High	How much of each sweep goes to Bluetooth
Sweep interval	3 / 5 / 8 / 12 s	Time between sweeps
Wi-Fi RSSI min / max	-100 to -10 dBm	Only log Wi-Fi heard inside this window (weaker than min or stronger than max is ignored)
BLE RSSI min / max	-100 to -10 dBm	The same for Bluetooth
GPS gate	Fix only / Always	Fix only: never sweep without a fix. (Every ride already waits for a fix with 4 or 5 satellites.)
GPS accuracy	Off / HDOP 5 / HDOP 2 / 6 sats	Wait for a more precise fix before sweeping
Skip open APs	Yes / No	Leave open (no password) networks out
Dedupe BSSID	On / Off	On: never log a network twice in a run. Off: log it again when it is 10 dB stronger.
Channels	All / 2.4 / 5 / 1-6-11	Which Wi-Fi channels to log
Live session	Open	Sweeps, last sweep time, unique per band, CSV rows, RSSI range, GPS, radios and log file
Exit Pro mode		Back to the normal screens. Your Pro settings are kept for next time.

10. Care, safety and responsible use

- **Driving.** Do not look at or touch the screen while driving. Set up and tap START before you move, or let a passenger use it. Mount the board so it cannot block your view, an airbag or the controls.
- **Know the law where you are.** Listening to openly broadcast Wi-Fi and Bluetooth is lawful in many places, but rules differ. Never join networks you are not allowed to use.
- **Respect people.** A log is a list of devices, not of people. Do not use it to identify, follow or harass anyone, and think before sharing raw logs that show where you live or work.
- **Heat.** A board on a dashboard in the sun gets very hot. Keep it shaded and take it with you when you park.
- **Power.** Use a proper USB supply or power bank. The board has no battery.
- **Your keys.** Your Wi-Fi password and API keys are stored in a plain file on the microSD card. Keep the card safe; if it is lost, change the keys in your WiGLE and WDGWars accounts.
- **Keep the software up to date** by flashing new releases (3.2).

Photosensitivity

A very small percentage of people may experience seizures or blackouts when exposed to certain light patterns or flashing lights, including on screen displays. This device shows animated graphics, moving elements and colour changes. If you or anyone in your family has an epileptic condition, consult a physician before use. Stop immediately if you feel dizziness, altered vision, eye or muscle twitches, loss of awareness, disorientation, any involuntary movement or convulsions. Use in a well-lit area and take regular breaks.

> **NOTE** No warranty
>
> SKIrf Wardriver is free software under the GNU General Public License v3 and comes with no warranty. You use it at your own risk.

11. Troubleshooting

The board prints helpful messages over USB: open any serial terminal at **115200 baud** on the port you flashed through. Every run start and stop is printed with the log file name.

What you see	What to try
The computer does not see the board, or flashing fails	USB-C data cable, left port, another USB port. Hold BOOT while plugging in. Unplug a buzzer while flashing.
Black screen	Unplug for 10 s and plug in again. Flash the .bin again.
Red and blue swapped everywhere	Your panel differs: build from source with SKIRF_PANEL_BGR 0 (docs/BUILDING.md) and please report it.
Taps land beside the target	Settings > Touch calibration. Cannot reach it? Hold BOOT while plugging in, or hold an empty spot on Home for 3 s.
Waiting for a GPS fix, for a long time	Go outdoors. Open GPS: SATELLITES should climb. Still 0? Check wiring and Settings > GPS pins (try the presets). The module must send NMEA at 9600 baud.
Waiting for more satellites	The fix is there but weak. Give the antenna a clearer view. In Pro mode set GPS accuracy to Off.
Few networks	Check the ride matches your speed. In Pro mode check Wi-Fi/BLE scan On, Channels All, Skip open APs No, wide RSSI windows, Sweep interval 3 s.
Settings shows Flash next to Logs	The card was not found: fit it before switching on, use FAT32 (32 GB or smaller), try another card.
Cannot reach the web portal	It is only up for 60 s. Join SKIrf_WARDRIVER, open http://192.168.4.1 (not https), turn off Wi-Fi assist on the phone.
"No internet WiFi" when uploading	Save your home Wi-Fi (6.5) and be in range of it. Stop any run first.
"No API key" when uploading	Save your WDGWars key in the web portal. For WiGLE use the API Name and API Token, not your password.
No sound	Fit a passive buzzer: + to IO26, - to GND.
Start again from scratch	Flash the .bin again: this resets the saved settings. Delete logs from the Logs screen or the portal.

Still stuck? Open an issue on the project page with the **Bug report** form. Include the release shown on the boot screen, what you did, and what the Serial Monitor printed.

Quick-reference card

Cut out or fold this page and keep it with the board.

START A RUN Home: START WARDRIVE. Tap your ride. Tap START. Wait for the GPS fix.	**STOP A RUN** Tap STOP, then TAP TO STOP within 3 s. The Run report follows.	**RIDES** CAR 2.5 s sweeps MOTO 3.5 s WALK/BIKE 6 s, every channel
STATUS DOT Green: GPS fix. Pale: satellites, no fix. Red blinking: recording.	**LOGS** Home: LOGS. Tap runs, ALL, WIGLE, WDGW. DELETE then CONFIRM.	**WEB PORTAL** Join SKIrf_WARDRIVER. Open http://192.168.4.1 60 s, or Settings > Web portal
UPLOAD KEYS WiGLE: API Name + Token from your account page. WDGWars: API key from your profile. Save in the portal.	**RECALIBRATE** Settings > Touch calibration, hold an empty spot on Home 3 s, or hold BOOT at power-on.	**GPS PINS** Default RX 4, TX 5. Settings > GPS pins, SAVE restarts.
FILE NAMES SKIrf_WD_C/M/W_date_time.csv WiGLE CSV 1.6	**FLASHING** Left USB-C port. write-flash 0x0 the .bin Hold BOOT if it will not connect.	**SOMETHING WRONG?** Serial Monitor 115200 baud. Section 11 of this guide.

Glossary

Term	Meaning
Wardriving	Moving around and recording the wireless networks you can hear, with their position, to map them.
SSID	The name of a Wi-Fi network.
BSSID / MAC	The hardware address of an access point or device. The first half identifies the maker.
Beacon	The short message an access point broadcasts several times a second to say it is there.
BLE	Bluetooth Low Energy: the kind of Bluetooth used by watches, trackers, speakers and sensors.
RSSI / dBm	Signal strength. -40 is very strong, -90 is very weak.
Sweep	One pass over every Wi-Fi channel on both bands, then Bluetooth.
Dwell	How long the board listens on each channel during a sweep.
DFS channels	5 GHz channels shared with radar. The board must listen passively there, which is slow.
Unique	Counted once per run, however often it is heard.
Fix	When the GPS knows where it is. HDOP is a measure of how precise the fix is (lower is better).
WiGLE	A worldwide map of wireless networks built from uploads by wardrivers (wigle.net).
WDGWars	A wardriving game with gangs and leaderboards (wdgwars.pl).
Web portal	The page the board serves on its own hotspot for 60 seconds.

Appendix A. Technical summary

Item	Detail
Board	NM-CYD-C5 (RockBase / NMTech), ESP32-C5
Display	2.8 inch ST7789, 240 x 320, SPI (SCK 6, MOSI 7, MISO 2, CS 23, DC 24), backlight IO25
Touch	XPT2046 resistive, CS IO1, shares the SPI bus
Radios	Wi-Fi 2.4 GHz + 5 GHz (country US by default: 5 GHz up to channel 165), Bluetooth LE
Scanning	ESP-IDF scan with per-ride channel dwell; asynchronous BLE scan; unique table of 16,384 entries in PSRAM (4,096 without PSRAM)
GPS	NMEA at 9600 baud on RX IO4 / TX IO5 by default; GGA + RMC at 5 Hz requested
Storage	microSD (FAT32, CS IO10, shared SPI bus) or LittleFS on the board's flash (about 1.5 MB)
Log format	WiGLE CSV 1.6, one file per run, rows written once per sweep
Web portal	Open hotspot SKIrf_WARDRIVER, http://192.168.4.1, 60 s; also on your home network while connected
Uploads	WiGLE (api.wigle.net) and WDGWars (wdgwars.pl) over HTTPS
Sound	Optional passive buzzer on IO26
Software	Firmware image for the ESP32-C5, SKIrf Wardriver Beta 1 (0.1.0-beta.1)

Appendix B. Log columns

Each file starts with a WiGLE pre-header line (app release, model, device) and then this header row:

Column	Contents
MAC	Hardware address of the network or device
SSID	Network name, or Bluetooth name when it has one
AuthMode	Security, for example [WPA2-PSK-CCMP][ESS]; Misc [LE] for Bluetooth
FirstSeen	GPS date and time (UTC)
Channel, Frequency	Wi-Fi channel and its frequency in MHz (blank for Bluetooth)
RSSI	Signal strength in dBm
CurrentLatitude, CurrentLongitude, AltitudeMeters	Where you were
AccuracyMeters	Estimated from the GPS HDOP
RCOIs, MfgrId	Roaming IDs (blank) and the Bluetooth manufacturer ID
Type	WIFI or BLE